PLANRS ACADEMY

COMPUTER SCIENCE & PROJECT-BASED LEARNING

Becoming a Full-Stack Creator: Building Your Capstone Project

Grade 6 Session: From ideation to architecture—learn how to combine your skills to build a real-world web app.

1. Be the CEO of Your Own App

Imagine you are the CEO of a brand-new Indian startup. You see a problem in your neighborhood—maybe people have extra home-cooked food to share, or your society needs a better way to track visitors. How do you build a solution that thousands of people can use?

This is the start of your **Capstone Project**. Over the next few weeks, you won't just be learning; you'll be **creating**. You are now a "Junior Software Architect."

The Visionary Mindset

Great apps like Zomato, Swiggy, and Paytm all started with a simple idea and a "Full-Stack" plan to make that idea work for everyone.

2. Understanding Full-Stack

A "Full-Stack" developer is someone who can build "The Whole Package"—both the parts people see and the parts that are hidden away.

The Frontend (The Face)

Everything the user interacts with—buttons, colors, images, and text. This is built using **HTML and CSS**.

The Backend (The Brain)

The hidden logic and storage. It handles the "thinking" and remembers information even when you close the app.

The Smart Home Analogy

The switches and beautiful lights you touch are the **Frontend**. The wiring and the main power box hidden inside the walls are the **Backend**.

3. Technologies Working Together

To build a modern app, we need all our previous skills to work as one team.

- **WebSockets:** Provide real-time updates, like watching a delivery partner move on a map.
- **Cloud Storage:** Keeping your data (like photos) safe in a "digital locker" like AWS or Azure.
- **Databases:** The digital "Godown" or warehouse where all your important information is kept organized.

The API Messenger

Think of an **API** as a waiter in a Dhaba. They take your order from the table (Frontend) to the kitchen (Backend) and bring your food back to you.

4. Your Digital Blueprint: MVC

Before building, every architect needs a blueprint. We use the **MVC Pattern** to keep our project organized.

View

The User Interface. What will your app look like? (The Plate)

Model

The Data. What information will your app save in the Database? (The Kitchen)

Controller

The Logic. How will your app respond when a user clicks a button? (The Waiter)

5. Finding Your Project Idea

The best apps solve real problems. Here are some ideas for your local Indian community:

Society Helper

Track gate entries, water deliveries, or manage common area bookings for your apartment complex.

Study Buddy

A real-time chat platform where classmates can help each other with homework in real-time.

Eco-Warrior

An app to log plastic recycling efforts in your colony and reward the biggest "Eco-Warriors."

Lassi Stall Tracker

A fun app to locate nearby lassi stalls, check their menus, and see special offers.

6. Indian Tech Success Stories

You are following in the footsteps of amazing creators who built apps that changed India.

- **Zomato & Swiggy:** They used Full-Stack tech to make sure you never have to go hungry.
- **Paytm:** They built a secure "digital vault" for money so we could go cashless.
- **MyGate:** They used simple "Society Helper" ideas to keep thousands of Indian homes safe.

All these companies started exactly where you are today: with an **idea** and a **blueprint**.

7. Activity: Drawing Your Blueprint

It's time to plan! Use these four questions to design your app's architecture.

The 4 Architecture Questions:

1. **View:** What will the users see on their screens? (Buttons, lists, maps?)
2. **Model:** What data will be saved in your "Digital Godown"? (Names, dates, points?)
3. **Real-Time:** Will it need live updates? (Do you need WebSockets?)
4. **Storage:** Will you save photos or big files in the Cloud?

Pro Tip: Keep it simple! Start with one main feature and make it work perfectly.

8. Summary: The Whole Package

Session Recap

1. **Full-Stack** combines the visible Face and the hidden Brain.
2. **MVC** is the blueprint that keeps your project organized.
3. You can use **Cloud, WebSockets, and Databases** as your tools.
4. Real-world apps solve **local community problems**.

You Are Ready!

Your journey as a Full-Stack Creator starts now. We can't wait to see what you build!